$E = N\varepsilon$

E :- Total energy of system
N :- Number of oscillators
ε :- Energy per oscillator

$$S = N\ln\Omega + \ln 2 - \ln(N-1)! + (N-1)\ln(N\varepsilon)$$

$$S = N\ln\Omega + \ln 2 - \ln(N-1)! + (N-1)\ln(N) + (N-1)\ln(\varepsilon)$$

$$\ln(N-1)! \approx (N-1)\ln(N-1) - (N-1)$$

For large N:
$\ln(N)! \approx N\ln N - N$

For large N: $\ln(N-1) \approx \ln N$

$$\ln(N-1)! \approx (N-1)\ln N - (N-1)$$

Sub in entropy:

$$S = N\ln\Omega + \ln 2 - [(N-1)\ln N - (N-1)] + (N-1)\ln N + (N-1)\ln\varepsilon$$

$$\Rightarrow S = N\ln\Omega + \ln 2 + (N-1) + (N-1)\ln\varepsilon$$

$$\Rightarrow S = N\ln\Omega + N + N\ln\varepsilon + \text{constants}$$

constants $= \ln 2 - 1 - \ln\varepsilon$, do not scale with N, ignored for large N

$$\Rightarrow S \approx N(\ln\Omega + 1 + \ln\varepsilon) + C$$

$$S \approx N(1 + \ln(\Omega\varepsilon)) + C$$

$$S \approx N s(\varepsilon) + \text{const} \quad (s(\varepsilon) = 1 + \ln(\Omega\varepsilon))$$

$S(\mathcal{E})$ is entropy for each harmonic oscillator

Originally Entropy was very messy
But after proper thermodynamic scaling
For large N.
All $N \ln N$ terms cancel
and only terms linear in N survive.

Constant terms $= \dfrac{\ln 2 - 1 - \ln \mathcal{E}}{N}$

will tend to zero as $N \to \infty$
Only terms linear in N survive

$\therefore \quad S \propto N$